BRITISH STANDARD

BS EN 15058:2006

Stationary source emissions — Determination of the mass concentration of carbon monoxide (CO) — Reference method: Non-dispersive infrared spectrometry

The European Standard EN 15058:2006 has the status of a British Standard

ICS 13.040.40

British Standards

National foreword

This British Standard is the official English language version of EN 15058:2006.

The UK participation in its preparation was entrusted by Technical Committee EH/2, Air quality, to Subcommitee EH/2/1, Stationary source emissions, which has the responsibility to:

— aid enquirers to understand the text;

— present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep UK interests informed;

— monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this subcommittee can be obtained on request to its secretary.

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English Version

Stationary source emissions — Determination of the mass concentration of carbon monoxide (CO) — Reference method: Non-dispersive infrared spectrometry

Emissions de sources fixes — Détermination de la concentration massique en monoxyde de carbone (CO) — Méthode de référence : spectrométrie infrarouge non dispersive

Emissionen aus stationären Quellen — Bestimmung der Massenkonzentration von Kohlenmonoxid (CO) — Referenzverfahren: Nicht-dispersive Infrarotspektrometrie

This European Standard was approved by CEN on 20 April 2006.

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Foreword

This document (EN 15058:2006) has been prepared by Technical Committee CEN/TC 264 “Air quality”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by November 2006, and conflicting national standards shall be withdrawn at the latest by November 2006.

This document has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

1 Scope

This European Standard specifies the Standard Reference Method (SRM) for sampling, and determining carbon monoxide content in ducts and stacks emitting to atmosphere. It describes the Non Dispersive Infra-Red (NDIR) analytical technique, including the sampling system and sample gas conditioning system, to determine CO in flue gases. This European Standard is the reference method for periodic monitoring and for calibration or control of Automatic Measuring Systems (AMS) permanently installed on a stack, for regulatory purposes or other purposes. To be used as the SRM, it is necessary to demonstrate that the performance characteristics of the method are lower than the performance criteria defined in this European standard and that the overall uncertainty of the method is less than ± 6 % relative at the daily Emission Limit Value (ELV).

NOTE When the NDIR method is used as an AMS, refer to EN 14181 and other relevant standards provided by CEN TC 264.

It is necessary for anybody who would like to use an Alternative Method instead of this Standard Reference Method to work out the demonstration of the equivalence according to the Technical Specification TS 14793, providing that his capability to carry out this demonstration is officially recognised by his national accreditation body or law.

This Standard Reference Method has been evaluated during field tests on waste incineration, co-incineration installations and large combustion plants. It has been validated for CO concentrations with sampling periods of 30 min in the range of 0 mg/m^3 to 400 mg/m^3 for large combustion plants and 0 mg/m^3 – 740 mg/m^3 for waste and co-incineration. For waste incineration plants, Council Directive 2000/76/EC lays down emission values which are expressed in mg/m^3, on dry basis at a specified value of O_2 and at reference conditions of 273 K and 101,3 kPa.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ENV 13005, *Guide to the expression of uncertainty in measurement*.

CEN/TS 14793: *Stationary source emission - Intralaboratory validation procedure for an alternative method compared to a reference method.*

EN ISO 14956, *Air Quality – Evaluation of the suitability of a measurement procedure by comparison with a required measurement uncertainty* (ISO 14956:2002).

ISO 5725-2, *Accuracy (trueness and precision) of measurement methods and results – Part 2: Basic method for the determination of* repeatability and reproducibility *of a standard measurement method*.

ISO 5725-6:1994, *Accuracy (trueness and precision) of measurement methods and results - Part 6: Use in practice of accuracy values*.

VIM 1994, *International vocabulary of basic and general terms in metrology*.

3 Terms and definitions

For the purposes of this document the following terms and definitions apply.

3.1
adjustment (of a measuring system)
operation of bringing a measuring system into a state of performance suitable for its use

[VIM 4.30]

3.2
ambient temperature
temperature of the air around the measuring system

3.3
automatic measuring system
(AMS)
measuring system permanently installed on site for continuous monitoring of emissions

NOTE 1 An AMS is a method which is traceable to a reference method.

NOTE 2 Apart from the analyser, an AMS includes facilities for taking samples (e.g. probe, sample gas lines, flow meters, regulators, delivery pumps) and for sample conditioning (e.g.dust filter, moisture removal devices, converters, diluters). This definition also includes testing and adjusting devices that are required for regular functional checks.

3.4
calibration
statistical relationship between values of the measurand indicated by the measuring system (AMS) and the corresponding values given by the standard reference method (SRM) used during the same period of time and giving a representative measurement on the same sampling plane

NOTE The result of calibration permits to establish the relationship between the values of the SRM and the AMS (calibration function).

3.5
drift
difference between two zero (zero drift) or span readings (span drift) at the beginning and at the end of a measuring period

3.6
emission limit value
(ELV)
emission limit value according to EU Directives on the basis of 30 min, 1 hour or 1 day

3.7
influence quantity
quantity that is not the measurand but that affects the result of the measurement

[adapted VIM 2.7]

EXAMPLES

— ambient temperature;

— atmospheric pressure;

— presence of interfering gases in the flue gas matrix;

— pressure of the gas sample.

3.8
interference
negative or positive effect upon the response of the measuring system, due to a component of the sample that is not the measurand

3.9
lack of fit
systematic deviation within the range of application between the measurement result obtained by applying the calibration function to the observed response of the measuring system measuring test gases and the corresponding accepted value of such test gases

NOTE 1 Lack of fit may be a function of the measurement result.

NOTE 2 The expression "lack of fit" is often replaced in everyday language by "linearity" or "deviation from linearity".

3.10
measurand
particular quantity subject to measurement

[VIM 2.6]

3.11
measuring system
complete set of measuring instruments and other equipment assembled to carry out specified measurements

[VIM 4.5]

3.12
performance characteristic
one of the quantities (described by values, tolerances, range…) assigned to equipment in order to define its performance

3.13
repeatability in the laboratory
closeness of the agreement between the results of successive measurements of the same measurand carried out under the same conditions of measurement

NOTE 1 Repeatability conditions include:

— same measurement procedure;

— same laboratory;

— same measuring system, used under the same conditions;

— same location;

— repetition over a short period of time.

NOTE 2 Repeatability may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the repeatability is expressed as a value with a level of confidence of 95 %.

[VIM 3.6]

3.14
repeatability in the field
closeness of the agreement between the results of simultaneous measurements of the same measurand carried out with two sets of equipment under the same conditions of measurement

NOTE 1 These conditions include:

- same measurement procedure;
- two sets of equipment, the performance of which fulfils the requirements of the reference method, used under the same conditions;
- same location;
- implemented by the same laboratory;
- typically calculated on short periods of time in order to avoid the effect of changes of influence parameters (e.g. 30 min).

NOTE 2 Repeatability may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the repeatability under field conditions is expressed as a value with a level of confidence of 95 %.

3.15
reproducibility in the field
closeness of the agreement between the results of simultaneous measurements of the same measurand carried out using several sets of equipment under the same conditions of measurement

NOTE 1 These conditions are called field reproducibility conditions and include:

- same measurement procedure;
- several sets of equipment, the performance of which fulfils the requirements of the reference method, used under the same conditions;
- same location;
- implemented by several laboratories.

NOTE 2 Reproducibility may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the reproducibility under field conditions is expressed as a value with a level of confidence of 95 %.

3.16
residence time in the measuring system
time period for the sampled gas to be transported from the inlet of the probe to the inlet of the measurement cell

3.17
response time
time interval between the instant when a stimulus is subjected to a specified abrupt change and the instant when the response reaches and remains within specified limits around its final steady value

NOTE By convention time taken for the output signal to pass from 0 % to 90 % of the final change.

[VIM 5.17]

3.18
sampling location
specific area close to the sampling plane where the measurement devices are set up

3.19
sampling plane
plane normal to the centreline of the duct at the sampling position

[EN 13284-1]

3.20
sampling point
specific position on a sampling line at which a sample is extracted

[EN 13284-1]

3.21
span gas
test gas used to adjust and check a specific point on the response line of the measuring system

NOTE This concentration is often chosen around 80 % of the upper limit of the range or around the emission limit value.

3.22
standard reference method
(SRM)
measurement method recognised by experts and taken as a reference by convention, which gives, or is presumed to give, the accepted reference value of the concentration of the measurand (3.10) to be measured

3.23
uncertainty
parameter associated with the result of a measurement, that characterises the dispersion of the values that could reasonably be attributed to the measurand

3.23.1
standard uncertainty
u
uncertainty of the result of a measurement expressed as a standard deviation u

3.23.2
expanded uncertainty
U
quantity defining a level of confidence about the result of a measurement that may be expected to encompass a specific fraction of the distribution of values that could reasonably be attributed to a measurand

$$U = k \times u$$

NOTE In this European Standard, the expanded uncertainty is calculated with a coverage factor of $k = 2$, and with a level of confidence of 95 %.

3.23.3
combined uncertainty
u_c
standard uncertainty u_c attached to the measurement result calculated by combination of several standard uncertainties according to GUM.

3.23.4
overall uncertainty
U_c
expanded combined standard uncertainty attached to the measurement result calculated according to GUM

$$U_c = k \times u_c$$

3.24
uncertainty budget
calculation table combining all the sources of uncertainty according to EN ISO 14956 or ENV 13005 in order to calculate the overall uncertainty of the method at a specified value

4 Principle

4.1 General

This European standard describes the reference method for sampling, and determining the carbon monoxide (CO) concentration in ducts and stacks emitting to atmosphere by means of an automatic analyser using Non Dispersive Infra-Red (NDIR) absorption principle. The specific components and the requirements for the sampling system and the NDIR analyser are described in Clause 6. A number of performance characteristics with associated minimum performance criteria are given for the analyser. These performance characteristics and the overall uncertainty of the method shall meet the performance criteria given in this European Standard. Requirements and recommendations for quality assurance and quality control are given for measurements in the field (see Table 1 in 7.2).

4.2 Measuring principle

CO concentration is measured with use of non-dispersive infrared methods. The attenuation of infrared light passing through a sample cell is a measure of the concentration of CO in the cell, according to the Lambert-Beer law. Not only CO but also most hetero-atomic molecules absorb infrared light, in particular water and CO_2 have broad bands that can interfere with the measurement of CO. Different technical solutions have been developed to suppress cross-sensitivity, instability and drift in order to design automatic monitoring systems with acceptable properties. For instance:

— measuring IR absorption of a specific wavelength (4,7 μm for CO);

— dual-cell monitors, using a reference cell filled with clean air (compensation for drift);

— gas filter correlation, "measuring" over a range of wavelengths.

Special attention shall be paid to infrared radiation absorbing gases such as water vapour, carbon dioxide, nitrous oxide and hydrocarbons.

NDIR analysers are combined with an extractive sampling system and a gas conditioning system. A representative sample of gas is taken from the stack with a sampling probe and conveyed to the analyser through the sampling line and gas conditioning system. The values from the analyser are recorded and/or stored by means of electronic data processing.

The concentration of carbon monoxide is measured in volume/volume units (if the analyser is calibrated using a volume/volume standard). The final results for reporting are expressed in milligrams per cubic meter using standard conversion factors (see Clause 10).

5 Sampling system

5.1 General

A representative volume (see 8.2) is extracted from the flue gas for a fixed period of time at a controlled flow rate. The sampling system consists of:

— sampling probe;

— filter;

— sampling line;

— conditioning system.

A filter removes the dust in the sampled volume before the sample is conditioned and passes to the analyser. Two different sampling and conditioning configurations can be used in order to avoid the water condensation in the measuring system. These configurations are:

— Configuration 1: removal of water vapour by condensation using a cooling system;

— Configuration 2: removal of water vapour through elimination using a permeation drier.

It is important that all parts of the sampling equipment upstream of the analyser are made of materials that do not react with or absorb CO. The temperature of its components coming into contact with the gas shall be maintained at a sufficiently high temperature to avoid any condensation and alter the gas composition.

Conditions and layout of the sampling equipment contribute to the overall uncertainty of the measurement. In order to minimise this contribution to the overall measurement uncertainty, performance criteria for the sampling equipment and sampling conditions are given in 5.2 and 7.2.

5.2 Sampling system

5.2.1 Sampling probe

In order to reach the representative measurement point(s) of the sampling plane, probes of different lengths and inner diameters may be used. The design and configuration of the probe used shall ensure the residence time of the sample gas within the probe is minimised in order to reduce the response time of the measuring system.

The procedure described in 8.2 shall be used when a lack of homogeneity in the flue gas is suspected.

5.2.2 Filter

The filter shall be made of an inert material (e. g. ceramic or sinter metal filter with an appropriate pore size). It shall be heated above the sample dew point temperature. The particle filter shall be changed or cleaned periodically depending on the dust loading at the sampling site.

NOTE Overloading of the particle filter may increase the pressure drop in the sampling line.

5.2.3 Sampling line

The sampling line shall be heated up to the conditioning system. It shall be made of a suitable corrosion resistant material (e. g. stainless steel, borosilicate glass, ceramic or titanium could be used; PTFE is only suitable for flue gas temperature lower than 200 °C).

NOTE Excessive temperature should be avoided because it might alter the flue gas characteristics.

5.2.4 Conditioning system

5.2.4.1 Sample cooler (configuration 1)

A dew-point temperature of 4 °C shall not be exceeded at the outlet of the sample cooler.

NOTE The concentrations, provided by this sampling configuration, are considered to be given on dry basis. However, the results may be corrected for the remaining water vapour (refer to the table of Annex A in EN 14790:2005).

5.2.4.2 Permeation drier (configuration 2)

The permeation drier is used before the gas enters the analyser in order to separate water vapour from the flue gas. A dew-point temperature of 4 °C shall not be exceeded at the outlet of the permeation drier.

NOTE The concentrations, provided by this sampling configuration, are considered to be given on dry basis. However, the results may be corrected for the remaining water vapour (refer to the table of Annex A in EN 14790:2005).

5.2.5 Sample pump

The sample pump shall be capable of operating to the specified flow requirements of the manufacturer of the analyser and pressure conditions required for the sample cell. The pump shall be resistant to corrosion. An external pump shall be consistent with the requirements of the analyser to which it is connected.

5.2.6 Secondary filter

The secondary filter is used to separate fine dust, with a pore size less than 5 µm. For example it may be made of glass-fibre, sintered ceramic, stainless steel or PTFE-fibre.

5.2.7 Flow controller and flow meter

This apparatus sets the required flow. A corrosion resistant material shall be used. The sample flow rate into the analyser shall be maintained within the analyser manufacturer's requirements. A controlled pressure drop across restrictors is usually employed to maintain flow rate control into the NDIR analyser.

NOTE No additional flow controller or flow meter is necessary when they are part of the analyser itself.

6 Analyser equipment

6.1 General

The main parts of the analyser are typically:

— source of infrared radiation;

— optics to focus the radiation through the measuring cell to the infrared detector;

— way of modulating the infrared beam;

— means to select a suitable wavelength or wavelengths to measure the gas;

— measuring cell that the sample gas enters. There may be a reference cell in some designs;

— infrared detector;

— amplifier and signal processing system to give an electrical output proportional to the CO concentration.

The standard of construction and vibration/corrosion resistance shall be suited to industrial environments and to the composition of the flue gas.

In Annex A schematic diagrams are given of two types of non-dispersive infrared analysers.

6.2 Pressure and temperature effects

The output signal of the analyser is proportional to the density of CO (number of CO molecules) present in the absorption cell and depends on the absolute pressure and temperature in the absorption cell. The effects of variations of pressure and temperature in the absorption cell should be taken into account by the manufacturer.

6.3 Sampling pump for the analyser

The sampling pump can be separate or part of the analyser. In any case, it shall be capable of operating within the specified flow requirements of the manufacturer of the analyser and pressure conditions required for the NDIR absorption cell.

6.4 Interferences due to infrared absorbing gases

6.4.1 General

As various gases absorb infrared radiation, interference from these gases can occur when their infrared absorption bands coincide or overlap the CO infrared absorption bands. The degree of interference varies among individual NDIR analysers. In general, gas correlation spectrophotometers are less sensitive to the influence of interferents.

6.4.2 Water vapour

The primary interferent is water vapour. However water vapour interference should be minimised by using sampling and conditioning configuration 1 or 2.

6.4.3 Other interferents

Other known interferents are carbon dioxide, hydrocarbons and N_2O. Knowledge of the gas composition and the cross sensitivity of the analyser is useful to ensure that none of the compounds interfere with the measurement.

7 Determination of the characteristics of the method: analyser, sampling and conditioning line

7.1 General

When this European standard is used as a reference method, the user shall demonstrate that:

— performance characteristics of the method given in Table 1 shall be equal or better than the associated performance criteria; and

— overall uncertainty calculated by combining values of standard uncertainties associated to the performance characteristics given in Table 1 is less than 6 % at the daily emission limit value, on dry basis and before correction to the specified value of O_2.

The values of the selected performance characteristics shall be evaluated by means of a laboratory test and a field test. An experienced laboratory recognised by the competent authority shall perform the laboratory tests. The user shall perform the field tests.

The user shall check the performance characteristics with a periodicity given in Table 4 (clause 9). These performance characteristics shall be determined using the appropriate CEN procedures.

NOTE This European standard does not give the list of relevant performance characteristics and criteria to check the conformity of an AMS.

7.2 Relevant performance characteristics of the method and performance criteria

The uncertainty of the measured values by the method is not only influenced by the performance characteristics of the analyser itself but also by:

— sampling line and conditioning system;

— site specific conditions;

— calibration gases used.

Table 1 gives an overview of the relevant performance characteristics and performance criteria, which shall be determined during laboratory and field tests according to the relevant CEN procedures, and indicates values included in the calculation of the overall uncertainty.

Table 1 — Relevant performance characteristics of the SRM

Performance characteristics	Lab. test for the analyser	Field test	Performance criterion	Performance characteristic to be included in calculation of overall uncertainty
Response time	X	X	≤ 200 s	
Detection limit	X		≤ ± 2,0 % of the range	
Lack of fit	X		≤ ± 2,0 % of the range	X
Zero drift	X		≤ ± 2,0 % of the range/24 h	X
Span drift	X		≤ ± 2,0 % of the range/24 h	X
Sensitivity to atmospheric pressure	X		≤ ± 3,0 % of the range/ 2 kPa	X
Sensitivity to sample volume flow of sample pressure	X		[a]	X
Sensitivity to ambient temperature	X		≤ ± 3,0 %of the range/10 K	X
Sensitivity to electric voltage	X		≤ ± 2,0 % of the range/10 V	X
Interferents[b]	X		Total ≤ ± 4,0 % of the range	X
Losses and leakage in the sampling line and conditioning system		X	≤ ± 2,0 % of the measured value	X
Standard deviation of repeatability in laboratory at zero	X		≤ ± 1,0 % of the range	X[c]
Standard deviation of repeatability in laboratory at span level	X		≤ ± 2,0 % of the range	X[c]

[a] The tested volume flow range or pressure is defined in the manufacturer's recommendations.

[b] Interferents that shall be tested are at least those given in Table 2.

The sums of contributions to uncertainty producing positive and negative effects are calculated separately. The maximum of their absolute value shall be compared with the performance criterion.

[c] Only one of these values shall be included in the calculation : the first possibility is to choose the repeatability standard deviation got from laboratory tests corresponding to the closest concentration to the actual concentration in stack, or the higher (relative) standard deviation of repeatability independently of the concentration measured in stack.

7.3 Establishment of the uncertainty budget

An uncertainty budget shall be established to determine if the analyser and its associated sampling system fulfils the requirements for a maximum allowable overall uncertainty.

The overall uncertainty for this method used as a reference shall be lower than 6 % at the daily emission limit value. This overall uncertainty is calculated on dry basis and before correction to the O_2 reference concentration.

The principle of calculation of the overall uncertainty is based on the law on propagation of uncertainty laid down in ENV 13005:

— determine the standard uncertainties for each value included in the calculation of the budget uncertainty by means of laboratory and field tests, and according to ENV 13005;

— calculate the uncertainty budget by combining all the standard uncertainties according to ENV 13005, including the uncertainty of the calibration gas and taking variations range of influence quantities and interferents of the specific site conditions into account. If these conditions are unknown, default values defined in Table 2 shall be applied. When corrections for residual water content in the flue gas are applied, the uncertainty attached to this correction shall be added to the uncertainty budget;

— values of standard uncertainty that are less than 5 % of the maximum standard uncertainty can be neglected;

— calculate the overall uncertainty at the daily emission limit value, on dry basis.

Table 2 — Default variations ranges of influence quantities and interferents to be applied for the determination of the uncertainty budget

Performance characteristic	Default variations range on site
Atmospheric pressure	± 2 kPa
Sample volume flow variation	In accordance with the manufacturer's recommendations
Ambient temperature	± 15 °C
Electric voltage at span level	230 V ± 20 V
CO_2	15 % volume
N_2O	40 mg/m^3
CH_4	57 mg/m^3
H_2O	1 % volume[a]

[a] This corresponds to a maximum concentration which is obtained by the sampling procedure described in 5.1.

An example of the evaluation of an uncertainty budget is given in Annex B.

8 Field operation

8.1 Sampling location

The sampling location is chosen to ensure that the gas concentrations measured are representative of the average conditions in the gas duct. In addition, the sampling location shall be chosen with regard to safety of the personnel, accessibility and availability of electrical power.

8.2 Sampling point

It is necessary to ensure that the gas concentrations measured are representative of the average conditions inside the waste gas duct. In most cases, a single sampling point situated in the middle of the duct shall be selected. For larger ducts, this point can be situated closer to the sampling port but not too close to avoid any disturbance of the flow or concentration due to influences from the sampling port.

However, when non-homogeneity of the flue gas is suspected, it shall be demonstrated if the flue gas is homogeneous or not. The homogeneity can be demonstrated with a continuous measurement of O_2 or CO_2 using a fixed and a moving probe. Sampling in non-homogeneous flue gases shall follow the requirements of relevant standards or technical specifications by CEN/TC 264.

8.3 Choice of the measuring system

To choose an appropriate analyser, sampling line and conditioning unit, the following characteristics of flue gases shall be considered before a field campaign:

— temperature of exhaust gases;

— flue gas moisture content and dew point;

— dust loading;

— expected concentration range of CO and emission limit values;

— expected concentration of potentially interfering substances, including at least the components listed in Table 2.

The full scale shall be adapted to the measuring task. Generally, this means that the scale is large enough to cover the peak emission and at least 150 % of the half hourly ELV.

To avoid long response times, the sample line should be as short as possible. If necessary a bypass pump should be used. Use an heated filter appropriate to the dust loading.

8.4 Setting of the analyser on site

8.4.1 General

The complete measuring system, the sampling line including the conditioning unit and the analyser, shall be connected according to the manufacturer's instructions. The probe is inserted so that its open end is at the representative point(s) in the duct (see 8.2).

After pre-heating, the flow passing through the sampling system and the analyser shall be adjusted to the chosen flow rate to be used during measurement. This flow should be maintained at a constant level.

The conditioning unit, sampling probe, filter, connection tube and analyser shall be stabilised with respect to temperature and humidity before adjustment of the analyser.

Time resolution of the data recording system shall be adapted to the measuring task and to the response time of the measuring system; generally, data is recorded at least every 60 s.

8.4.2 Preliminary zero and span check, and adjustments

8.4.2.1 Test gases

The zero gas shall be a gas containing no significant amount of carbon monoxide (For example, nitrogen or purified air).

The span gases used to adjust the analyser shall have a certified concentration of CO. The uncertainty on the analytical certificate of the span gas shall be less than or equal to ± 2 % for CO.

When the analyser is used for regulatory purposes, the span gas shall have a known concentration of approximately the half hourly ELV, or 50 % to 90 % of the selected range of the analyser.

8.4.2.2 Adjustment of the analyser

At the beginning of the measuring period, zero and span gases are supplied to the analyser directly, without passing through the sampling system. Adjustments are made until the correct zero and span gas values are given by the data sampling system:

- adjust the zero value;
- adjust the span;
- finally, check again zero to see if there is no significant changes (zero deviation lower than 2 times the repeatability at zero). If there is any problem, repeat the procedure.

8.4.2.3 Check of the sampling system including the leak test

Zero and span gas are supplied to the analyser through the sampling system, as close as possible to the tip of the line (in front of the filter if possible). Differences between the readings obtained during the adjustment of the analyser and during the check shall be lower than 2 % rel.

Additionally to this, the sampling line can be checked for leakage according to the following procedure or any other relevant procedure.

Assemble the complete sampling system. Seal off the end of the probe and switch on the pump. After reaching minimum pressure, read or measure the flow rate with an appropriate measuring device. The leak flow rate shall not exceed 2 % of the expected sample gas flow rate used during measurement.

8.4.3 Zero and span checks after measurement

At the end of the measuring period and at least once a day, zero and span checks shall be performed at the inlet of the sampling system by supplying test gases. The information shall be documented. In case of deviation between checks before and after measurement, values of deviation shall be indicated in the report.

If the span or zero drifts are higher than 2 % of the span value, it is necessary to correct both for zero and span drifts (see in Annex D a procedure of correction of data for drift effect). The results shall be rejected if the drift in zero or span is higher than 5 %.

9 Ongoing quality control

9.1 Introduction

Quality control is critically important in order to ensure that the uncertainty of the measured values for carbon monoxide is kept within the stated limits during extended automatic monitoring periods in the field. This means that maintenance, as well as zero and span adjustment procedures shall be followed, as they are essential for obtaining accurate and traceable quality data.

9.2 Frequency of checks

Table 4 shows the minimum required frequency of checks. The laboratory shall implement the relevant European Standards for determination of performance characteristics.

Table 3 — Frequency of checks as reference method

Checks	Frequency	Action criteria
Cleaning or changing of particulate filters[a] at the sampling inlet and at the monitor inlet	Every campaign if needed[a]	
Regular maintenance of several parts of the monitor	As required by manufacturer	As required
Lack of fit	At least every year and after repair of the analyser	As required and when lack of fit > ± 2 % of the range

[a] The particle filter shall be changed periodically depending on the dust loading at the sampling site. During this filter change the filter housing shall be cleaned. Overloading of the particle filter may increase the pressure drop in the sampling line.

10 Expression of results

The readings from the analyser are converted to concentrations using the appropriate calibration graph and the results are expressed in milligrams per cubic metre, or parts per million by volume, on dry basis at 273 K and 101,3 kPa.

For carbon monoxide (CO) the conversion factors at 273 K and 101,3 kPa are:

— 1 mg CO/m^3 = 0,8 ppm (V/V) CO.

— 1 ppm (V/V) CO = 1,25 mg CO/m^3.

— When results are given on wet basis, the concentrations on dry basis are calculated according to the following equation, where "H_2O" is the water vapour content determined according to [3]:

$$C_d = \frac{100}{100 - H_2O} C_w \tag{1}$$

Where

C_d is the concentration of the measurand on dry basis;

C_w is the measured concentration of the measurand on wet basis;

H_2O is the water vapour content measured in gas, in percentage/volume.

Concentrations are generally expressed at a reference concentration of O_2 defined in the European directives:

— 11 % for incineration of waste in waste incinerators;

— 10 % for co-incineration of waste in cement kilns;

— 3 % for combustion of gas or liquid fuels;

— 6 % for combustion of solid fuels;

— 15 % for gas turbines.

The corrected concentration of measurand C_{corr} is calculated using the followed equation:

$$C_{corr} = \frac{21 - O_{2,\ ref}}{21 - O_{2,\ mes}} C_{actual} \qquad (2)$$

Where

C_{corr} is the corrected concentration of measurand;

C_{actual} is the measured concentration of measurand at actual O_2 concentration;

$O_{2,\ mes}$ is the measured mean dry oxygen content during the sampling time;

$O_{2,\ ref}$ is the oxygen reference concentration.

11 Evaluation of the method in the field

The method has been evaluated during six field tests, on waste incineration installations, co-incineration installations and large combustion plants. Each test was performed by at least four different European measuring teams originating from ten CEN member countries.

The characteristics of installations, the conditions during field tests and the values of repeatability and reproducibility in the field are given in Annex C.

12 Equivalency with an alternative method

In order to show that an alternative method is equivalent to the reference method according to the CEN/TS 14793, the maximum allowable standard deviation of repeatability and standard deviation of reproducibility shall be calculated according to the following equations worked out from data got from the six validation field tests, the results of which are presented in Annex C.

The maximum allowable standard deviation of repeatability and reproducibility expressed in mg/m^3 for this reference method is:

$$S_{r\ limit}\ (C) = S_R\ (C) = 0{,}0118\, C + 3 \qquad (3)$$

where

C is the concentration in milligrams per cubic meter.

13 Test report

The test report shall include the following information:

a) reference to this European standard;

b) description of the purpose of tests;

c) information about the personnel involved in the measurement (name, qualification, etc.);

d) principle of gas sampling and conditioning;

e) information about the analyser and description of the sampling and conditioning line;

f) identification of the analyser used, and analyser's performance characteristics as listed in Table 1;

g) operating range;

h) details of the quality and the concentration of the calibration gases used;

i) details on the adjustment performed before and after actual sampling (at the inlet of the sampling line and at the inlet of the analyser);

j) description of plant and process;

k) identification of the sampling plane;

l) description of the location of the sampling point(s) in the sampling plane;

m) description of the operating conditions of the plant process;

n) changes in the plant operations during sampling, for example burner changes;

o) measurement results with sampling date, time and duration;

p) information on flue gas characteristics (temperature, velocity, moisture, pressure);

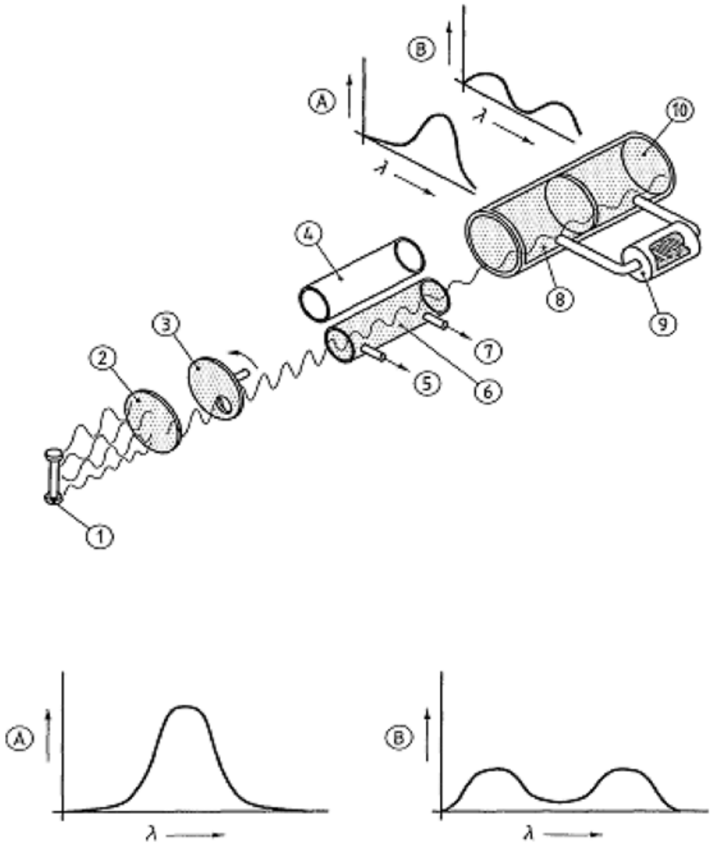
q) time averaging on relevant periods;

r) uncertainty budget established according to EN ISO 14956 or ENV 13005;

s) any deviations from this European standard.

Annex A
(informative)

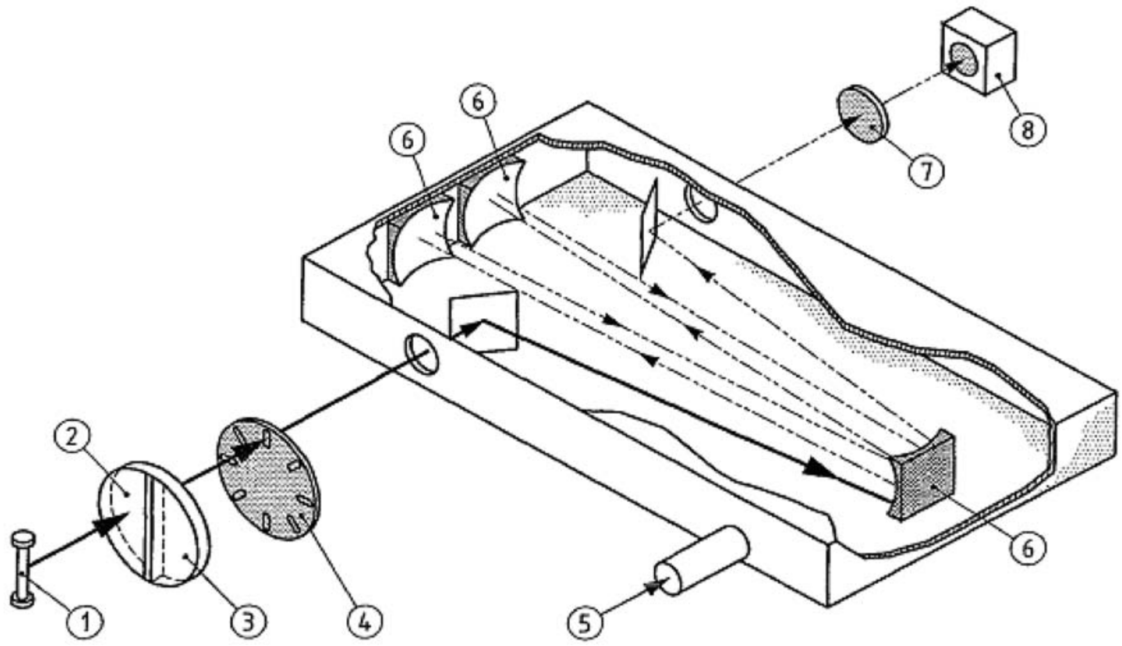
Schematics of non-dispersive infrared spectrometer



Key

1 IR lamp
2 Light filter
3 Chopper
4 Reference cell
5 Gas in
6 Sample cell
7 Gas out
8 First chamber
9 Microflow sensor
10 Second chamber
A Light absorption in first chamber
B Light absorption in second chamber
λ Wavelength

Figure A.1 — Example of a dual cell analyser



Key

1 IR source
2 Neutral Filter (N_2)
3 Gas filter cell (CO)
4 Modulator
5 Sample gas
6 Mirror
7 Filter
8 Detector

Figure A.2 — Example of a gas filter correlation analyser

Annex B
(informative)

Example of assessment of compliance of non-dispersive infrared method for CO with requirements on emission measurements

B.1 Process of uncertainty estimation

This informative annex gives an example of calculation of uncertainty budget established with configuration 1.

B.2 Process of uncertainty estimation

The procedure for calculating measurement uncertainty as follows is based on the law on propagation of uncertainty laid down in ISO 14956 or ENV 13005. The calculation procedure presents different steps.

B.2.1 Determination of model function

Define the measurand and all the parameters that influence the result of the measurement. These parameters, called “input quantities” shall be clearly defined.

Identify all sources of uncertainty contributing to any of the input quantities or to the measurand directly.

Then the model function, that is to say the relationship between the measurand and the influence quantities, shall be established, if possible in mathematical equation form.

B.2.2 Quantification of uncertainty components

Each uncertainty source is estimated to obtain its contribution to the overall uncertainty.

Use available performance characteristics of the measurement system, data from the dispersion of repeated measurements, data provided in calibration certificates…

Convert all uncertainty components (e. g. performance characteristics) to standard uncertainties of input and influence quantities.

B.2.3 Calculation of the combined uncertainty

Then the combined uncertainty u_{c} is calculated by combining standard uncertainties, by applying the “law of propagation of uncertainty”.

In general, the uncertainty associated to a concentration is expressed in overall uncertainty form. The overall uncertainty U_{c} corresponds to the expanded combined uncertainty, obtained by multiplying by a coverage factor “k”: $U_{\mathrm{c}} = k \times u_{\mathrm{c}}$. The value of the coverage factor k is chosen on the basis of the level of confidence required. In most cases, k is taken equal to 2, for a level of confidence of approximately 95 %.

The following Tables give one example of:

- specific conditions of the site (Table B.1), that is to say the values and the variation range of the influence parameters;
- performance characteristics of the method (Table B.2) related to the parameters which can have an influence on the response of the analyser: the metrological performances of the analyser and the effect of influence parameters (environmental conditions like ambient temperature, voltage, pressure and chemical interferents).

B.3 Specific conditions in the site

Table B.1 — Site specific conditions, applied for the example

Specific conditions	Value/range
Range of analyser	0 ppm to 100 ppm (0 mg/m^3 to 125 mg/m^3)
Studied concentration of CO (limit value of CO for the site)	50 mg/m^3 (n) corresponding to 40 ppm at $O_{2, ref}$
Conditions on the field Sample volume flow Temperature during adjustment Fluctuations of ambient temperature during meas- Voltage variation Atmospheric pressure during adjustement Atmospheric pressure variation	 60 l/h ± 5 l/h 285 K 283 K to 308 K 230 V ± 5 % 99 kPa 99 kPa to 100 kPa
O_2 reference concentration: $O_{2, ref}$	11 % volume
CO_2 concentration variations in the field N_2O concentration variations in the field CH_4 concentration variations in the field	8 % to12 % negligible 0 mg/m^3 to 10 mg/m^3
Calibration gas CO in N_2, without interferent	80 ppm ± 2 %

B.4 Performance characteristics of the method

Table B.2 — Performance characteristics, applied for the example

Performance characteristic for in-fra-red method	Performance criteria	Example of results laboratory and field tests
Response time	≤ 200 s	120 s
Detection limit	≤ ± 2 % of the range	± 0,6 % of range
Lack of fit	≤ ± 2 % of the range	± 0,6 % of range
Zero drift	≤ ± 2 % of the range/24 h	± 0,01 % of range/24 h
Span drift	≤ ± 2 % of the range/24 h	± 0,5 % of range/24 h
Sensitivity to the sample volume flow	≤ 1 % of the range	0,2 % of the range/10 l/h
Sensitivity to atmospheric pressure	≤ 3 % of the range/2 kPa	0,4 % of the measured value/kPa
Sensitivity to ambient temperature	≤ 3 % of the range/10 K	0,5 % of range/10 K
Sensitivity to electric voltage at span level	≤ 2 % of the range/10 V	0,12 % of range/10 V
Interferents CO_2 (15 %) N_2O (20 mg/m^3) CH_4 (50 mg/m^3)	Total : ≤ ± 4 % of the range	 -0,6 ppm 0,8 ppm 1,6 ppm
Standard deviation of repeatability in laboratory at zero	≤ 1 % of the range	0,3 % of range
Standard deviation of repeatability in laboratory at span level	≤ 2 % of the range	0,45 % of range

B.5 Calculation of standard uncertainty of concentration values given by the analyser

B.5.1 General

Model equations in this clause, as well as calculations of partial uncertainties are related to the values measured by the analyser and expressed in ppm.

Uncertainty associated to conversions in mg/m^3(n) is handled later in B.6.

B.5.2 Model equation and application of rule of uncertainty propagation

The concentration in CO is equal to the concentration given by the analyser plus corrections due to deviations associated to influence quantities and to the performance characteristics of the analyser.

$$C_{\mathrm{CO,pp}m} = C_{\mathrm{CO,read}} + Corr_{\mathrm{fit}} + Corr_{\mathrm{0,\,dr}} + Corr_{\mathrm{s,\,dr}} + Corr_{\mathrm{rep}} + Corr_{\mathrm{adj}} + \sum_{j=1}^{p} Corr_{\mathrm{inf}} + Corr_{\mathrm{int}} \quad \text{(B.1)}$$

Where

$C_{\mathrm{CO,\,ppm}}$	is the concentration of CO in ppm;
$C_{\mathrm{CO,\,read}}$	is the concentration of CO given by the analyser;
$Corr_{\mathrm{fit}}$	is the correction of lack of fit;
$Corr_{\mathrm{0,\,dr}}$, $Corr_{\mathrm{s,\,dr}}$	are the corrections of zero drift and span drift;
$Corr_{\mathrm{rep}}$	is the correction of repeatability of the measurement;
$Corr_{\mathrm{adj}}$	is the correction of adjustment;
$\sum_{j=1}^{p} Corr_{\mathrm{inf}}$	is the correction of influence quantities (ambient temperature, atmospheric pressure, sample volume flow, voltage);
$Corr_{\mathrm{int}}$	is the correction of interferents.

The corrections are specific for each analyser. Sometimes, value of correction is nil. For example, influence parameters (influence quantities: ambient temperature, atmospheric pressure and interferents), can increase or decrease during the measurement period and are generally not monitored; therefore, we can consider that the best correction to be applied is zero.

Whether the corrections are equal to zero or not, the uncertainties associated to these corrections shall be taken into account in the calculation of the global uncertainty. The combined standard uncertainty associated to CO measurement is given by:

$$u^2(C_{\mathrm{CO,ppm}}) = \sum_{i=1}^{N} \left[\left(\frac{\partial C_{\mathrm{CO,\,ppm}}}{\partial Xi} \right)^2 u^2(Xi) \right] \quad \text{(B.2)}$$

Where

$u(C_{\text{CO, ppm}})$	is the combined standard uncertainty associated to CO measurement;
Xi, I = 1 to N	are the influence quantities;
$\frac{\partial C_{\text{CO,ppm}}}{\partial Xi}$	is the sensitivity coefficient of $C_{\text{CO,ppm}}$, to X_i ;
$u(Xi)$	is the standard uncertainty associated to X_i ;
$\frac{\partial C_{\text{CO,ppm}}}{\partial Xi} u(Xi)$	is the partial uncertainty related to X_i .

The development of equation (B.2) gives:

$$\begin{aligned} u^2(C_{\text{CO,ppm}}) = & \left(\frac{\partial C_{\text{CO,ppm}}}{\partial C_{\text{CO,read}}}\right)^2 u^2(C_{\text{CO,read}}) + \left(\frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{fit}}}\right)^2 u^2(Corr_{\text{fit}}) + \left(\frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{0,\text{dr}}}\right)^2 u^2(Corr_{0,\text{dr}}) \\ & + \left(\frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{s,dr}}}\right)^2 u^2(Corr_{\text{s,dr}}) + \left(\frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{rep}}}\right)^2 u^2(Corr_{\text{rep}}) + \left(\frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{adj}}}\right)^2 u^2(Corr_{\text{adj}}) \\ & + \left(\frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{temp}}}\right)^2 u^2(Corr_{\text{temp}}) + \left(\frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{a,press}}}\right)^2 u^2(Corr_{\text{a,press}}) + \left(\frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{s,vf}}}\right)^2 u^2(Corr_{\text{s,vf}}) \\ & + \left(\frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{V}}}\right)^2 u^2(Corr_{\text{V}}) + \left(\frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{int}}}\right)^2 u^2(Corr_{\text{int}}) \end{aligned} \tag{B.3}$$

Definitions are given in Table 5.3.

B.5.3 Calculation of the partial uncertainties

— Sensitivity coefficients

$$\frac{\partial C_{\text{CO,ppm}}}{\partial C_{\text{CO,read}}} = \frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{fit}}} = \frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{0,\text{dr}}} = \frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{s,dr}}} = \frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{rep}}} = \frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{adj}}} = \frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{temp}}} = \frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{a,press}}}$$
$$= \frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{s,vf}}} = \frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{V}}} = \frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{int}}} = 1$$

— $u(C_{\text{CO,read}})$: the uncertainty related to the reading of the concentration is due to the resolution of the analyser and of the data acquisition. The uncertainty can be considered as negligible.

— $u(Corr_{\text{fit}})$: if $X_{\text{fit,max}}$ is the performance characteristic equal to the maximum deviation between the measured value and the value given through the linear regression achieved during the laboratory test, then, we can consider that the lack of fit has an equal probability to take any value included in the interval [-$X_{\text{fit,max}}$; + $X_{\text{fit,max}}$]. The standard uncertainty is calculated by application of a rectangular probability distribution.

If the performance characteristic is expressed in % of the range:

$$u(Corr_{\text{fit}}) = \frac{X_{\text{fit,max}} / 100 \times \text{range}}{\sqrt{3}}$$

$$\boxed{\frac{\partial C_{CO,ppm}}{\partial Corr_{fit}} u(Corr_{fit}) = \frac{X_{fit,max}/100 \times \text{Range}}{\sqrt{3}}}$$

— $u(Corr_{0,dr})$, $u(Corr_{s,dr})$: where $X_{0,dr}$ and $X_{s,dr}$ are the zero and the span drift, we can suppose that the drifts have the same probability to take any value included in the intervals [- $X_{0,dr}$;+ $X_{0,dr}$] and [- $X_{s,dr}$;+ $X_{s,dr}$]. The standard uncertainties are calculated by application of a rectangular probability distribution:

$$u(Corr_{0,dr}) = \frac{X_{0,dr}}{\sqrt{3}} \text{ and } u(Corr_{s,dr}) = \frac{X_{s,dr}}{\sqrt{3}}$$

Then $\boxed{\frac{\partial C_{CO,ppm}}{\partial Corr_{0,dr}} u(Corr_{0,dr}) = \frac{X_{0,dr}}{\sqrt{3}} \quad \text{and} \quad \frac{\partial C_{CO,ppm}}{\partial Corr_{s,dr}} u(Corr_{s,dr}) = \frac{X_{s,dr}}{\sqrt{3}}}$

— $u(Corr_{rep})$: the standard uncertainty due to the repeatability is equal to the repeatability standard deviation calculated from the results of the repetitions of the measurements.

Several tests can be carried out at different concentrations (at least at zero and at one span level). But only one of the values shall be included in the calculation:

— choose the repeatability standard deviation corresponding to the closest concentration measured in stack (in the example: the emission limit value);

— or the highest repeatability (relative) standard deviation whatever is the concentration measured in stack.

In the example, the highest repeatability standard deviation has been chosen:

$$u(Corr_{rep}) = \max(S_{0,rep} ; S_{s,rep}) = S_{rep}$$

Where

$S_{0,rep}$ is the standard uncertainty at level zero;

$S_{s,rep}$ is the standard uncertainty at span level.

Then $\boxed{\frac{\partial C_{CO,ppm}}{\partial Corr_{rep}} u(Corr_{rep}) = S_{rep}}$

— Standard uncertainties associated to the influence quantities $u(Corr_{temp})$, $u(Corr_{a,press})$, $u(Corr_{s,vf})$, $u(Corr_{V})$: during laboratory test, the influence quantities are tested for one value of the parameter and the effects of the influence quantities are supposed to be proportional to the value of the parameter. The correction of the effect of an influence quantity is also proportional to its variation.

If X_j is an influence quantity: $Corr_{Xj} = c_j \times \Delta X_j$

Where

c_j is a constant;

ΔX_j is the variation of the influence quantity.

The uncertainty associated to the correction is given by

$$u^2(Corr_{Xj}) = \left(\frac{\partial Corr_{Xj}}{\partial \Delta X_j}\right)^2 u^2(\Delta X_j)$$

Where

$\frac{\partial Corr_{Xj}}{\partial \Delta X_j} = c_j$ is the sensitivity coefficient; the sensitivity factor corresponds to the effect of the variation ΔX_j of the influence quantity in the response of the analyser. Sensitivity coefficients are determined by laboratory tests;

$u(\Delta X_j)$ is the standard uncertainty associated to the variation of the influence quantity between the calibration period and the measuring period. We can suppose that the influence quantity has the same probability to take any value included in the interval ΔX_j.

In the standard EN ISO 14956, the calculation of the standard uncertainty associated to the variation of the influence quantity is given by the equation:

$$\boxed{u(\Delta Xj) = \sqrt{\frac{(x_{j,max} - x_{j,adj})^2 + (x_{j,min} - x_{j,adj}) \times (x_{j,max} - x_{j,adj}) + (x_{j,min} - x_{j,adj})^2}{3}}}$$

where

$x_{j,min}$ is the minimum value of the influence quantity X_j during the measuring period;

$x_{j,max}$ is the maximum value of the influence quantity X_j during the measuring period;

$x_{j,adj}$ is the value of the influence quantity X_j during the adjustment of the analyser.

Then $\frac{\partial C_{CO,\ ppm}}{\partial Corr_{Xj}} u(Corr_{Xj}) = \frac{\partial Corr_{Xj}}{\partial X_j} u(\Delta X_j)$

$$\boxed{\frac{\partial C_{CO,\ ppm}}{\partial Corr_{Xj}} u(Corr_{Xj}) = c_j \sqrt{\frac{(x_{j,max} - x_{j,adj})^2 + (x_{j,min} - x_{j,adj}) \times (x_{j,max} - x_{j,adj}) + (x_{j,min} - x_{j,adj})^2}{3}}}$$

The calculation can be simplified in the two following cases:

— First case: if the value $X_{j,adj}$ is at the centre of the interval $\lfloor X_{j,max} ; X_{j,min} \rfloor$:

$$u(\Delta X_j) = \frac{(x_{j,max} - x_{j,min})}{\sqrt{12}} \text{ and } \boxed{\frac{\partial C_{CO,ppm}}{\partial Corr_{Xj}} u(Corr_{Xj}) = c_j \frac{(x_{j,max} - x_{j,min})}{\sqrt{12}}}$$

— Second case: if the value $X_{j,adj}$ is equal to $X_{j,max}$ or $X_{j,min}$:

$$u(\Delta X_j) = \frac{(x_{j,max} - x_{j,min})}{\sqrt{3}} \text{ and } \boxed{\frac{\partial C_{CO,ppm}}{\partial Corr_{Xj}} u(Corr_{Xj}) = c_j \frac{(x_{j,max} - x_{j,min})}{\sqrt{3}}}$$

— Standard uncertainties associated to the interferents: the case of the interferents is similar to the case of influence quantities. The effect of the interferents is tested for one concentration of interferent and is supposed to be proportional to the value of the parameter. The correction of the effect of an influence quantity is also proportional to its variation.

If Int_j is an interferent for the measured compound, and ΔInt_j its variation interval during the measurement period, the standard uncertainty of the correction is given by

$$u^2(Corr_{\mathrm{Intj}}) = \left(\frac{\partial Corr_{\mathrm{Intj}}}{\partial \Delta Int_j}\right)^2 u^2(\Delta Int_j)$$

where

$\frac{\partial Corr_{\mathrm{Intj}}}{\partial \Delta Int_j}$ is the sensitivity coefficient; sensitivity coefficients are determined by laboratory tests;

$u(\Delta Int_j)$ is the standard uncertainty associated to the variation of the concentration of the interferent Int_j during the measuring period. We can suppose that the influence quantity has the same probability to take any value included in the interval ΔInt_j.

In the standard EN ISO 14956, the calculation of the standard uncertainty associated to the variation of the influence quantity is given by the equation:

$$u(\Delta Int_j) = \sqrt{\frac{(Int_{j,\mathrm{max}} - Int_{j,\mathrm{adj}})^2 + (Int_{j,\mathrm{min}} - Int_{j,\mathrm{adj}}) \times (Int_{j,\mathrm{max}} - Int_{j,\mathrm{adj}}) + (Int_{j,\mathrm{min}} - Int_{j,\mathrm{adj}})^2}{3}}$$

where

$Int_{j,\mathrm{min}}$ is the minimum concentration of the interferent Int_j during the measuring period;

$Int_{j,\mathrm{max}}$ is the maximum concentration of the interferent Int_j during the measuring period;

$Int_{j,\mathrm{adj}}$ is the concentration of the interferent Int_j in the calibration gas used to adjust the analyser.

Then $\frac{\partial C_{\mathrm{CO,ppm}}}{\partial Corr_{\mathrm{Intj}}} u(Corr_{\mathrm{Intj}}) = \left(\frac{\partial Corr_{\mathrm{Intj}}}{\partial \Delta Int_j}\right) u(\Delta Int_j)$

$$\boxed{\frac{\partial C_{\mathrm{CO,ppm}}}{\partial Corr_{\mathrm{Intj}}} u(Corr_{\mathrm{Intj}}) = \frac{c_j}{Int_{j,\mathrm{test}}} \times \sqrt{\frac{(Int_{j,\mathrm{max}} - Int_{j,\mathrm{adj}})^2 + (Int_{j,\mathrm{min}} - Int_{j,\mathrm{adj}}) \times (Int_{j,\mathrm{max}} - Int_{j,\mathrm{adj}}) + (Int_{j,\mathrm{min}} - Int_{j,\mathrm{adj}})^2}{3}}}$$

Where

c_j is the sensitivity coefficient;

$Int_{j,test}$ is the concentration of the interferent used to determine the sensitivity coefficient c_j during laboratory test

$$\frac{c_j}{Int_{j,test}} = \frac{\partial Corr_{Intj}}{\partial \Delta Int_j}$$

The calculation can be simplified if the value $X_{j,adj}$ is equal to zero, that is to say that the concentration of the interferent in the calibration gas is equal to zero, what is often the case. The standard uncertainty is given by:

$$u(\Delta Int_j) = \sqrt{\frac{Int_{j,max}{}^2 + Int_{j,min} \times Int_{j,max} + Int_{j,min}{}^2}{3}}$$

$$\boxed{\frac{\partial C_{CO\,ppm}}{\partial Corr_{Intj}} u(Corr_{Intj}) = \frac{c_j}{Int_{j,test}} \times \sqrt{\frac{Int_{j,max}{}^2 + Int_{j,min} \times Int_{j,max} + Int_{j,min}{}^2}{3}}}$$

All interferents shall not be included in the calculation of the global uncertainty. In accordance with the standard ISO 14956, the calculation shall include the highest value between the sum $S_{int,\,p}$ of the interferents with positive impact in the response of the analyser and the sum $S_{int,\,n}$ of the interferents with negative impact.

$$S_{int,p} = \frac{\partial C_{CO,ppm}}{\partial Corr_{Int,p1}} u(Corr_{int,p1}) + \frac{\partial C_{CO,ppm}}{\partial Corr_{Int,p2}} u(Corr_{int,p2}) + ... + \frac{\partial C_{CO,ppm}}{\partial Corr_{Int,pk}} u(Corr_{int,pk})$$

$$S_{int,n} = \frac{\partial C_{CO,ppm}}{\partial Corr_{Int,n1}} u(Corr_{int,n1}) + \frac{\partial C_{CO,ppm}}{\partial Corr_{Int,n2}} u(Corr_{int,n2}) + ... + \frac{\partial C_{CO,ppm}}{\partial Corr_{Int,nt}} u(Corr_{int,nt})$$

Where

$S_{int,p}$ is the sum of the interferents with positive impact;

$\frac{\partial C_{CO,ppm}}{\partial Corr_{Int,p1}} u(Corr_{int,p1})$ is the partial uncertainty due to the 1st interferent with positive impact;

$\frac{\partial C_{CO,ppm}}{\partial Corr_{Int,pk}} u(Corr_{int,pk})$ is the partial uncertainty due to the k^{th} interferent with positive impact;

$S_{int,n}$ is the sum of the interferents with negative impact;

$\frac{\partial C_{CO,ppm}}{\partial Corr_{Int,n1}} u(Corr_{int,n1})$ is the partial uncertainty due to the 1st interferent with negative impact;

$\frac{\partial C_{CO,\,ppm}}{\partial Corr_{Int,nt}} u(Corr_{int,nt})$ is the partial uncertainty due to the t^{th} interferent with negative impact;

$$\boxed{\left(\frac{\partial C_{\text{CO,ppm}}}{\partial Corr_{\text{int}}}\right)^2 u^2(Corr_{\text{int}}) = \max\left[S_{\text{int,p}}{}^2; S_{\text{int,n}}{}^2\right]}$$

— $u(Corr_{\text{adj}})$: the standard uncertainty $u(Corr_{\text{adj}})$ is calculated from the uncertainty of the calibration gas. In general the uncertainty given by manufacturer is the expanded uncertainty; if X_{cal} is the expanded uncertainty of the calibration gas expressed in % relative, the standard uncertainty of the correction at ELV level is:

$$u(Corr_{\text{adj}}) = \frac{X_{\text{cal}}/100 \times ELV}{2}$$

Then $\boxed{\frac{\partial C_{\text{COppm}}}{\partial Corr_{\text{adj}}} u(Corr_{\text{adj}}) = \frac{X_{\text{cal}}/100 \times ELV}{2}}$

B.5.4 Result of combined uncertainty calculation

Table B.3 — Results of partial uncertainties calculation

Performance characteristic	Partial standard uncertainty	Value of partial standard uncertainty at limit value (in ppm)
Lack of fit	$u(Corr_{\text{fit}})$	$\frac{(0{,}6/100)\text{x}\,100}{\sqrt{3}} = 0{,}35$
Zero drift	$u(Corr_{\text{0,dr}})$	$\frac{(0{,}01/100)\text{x}\,100}{\sqrt{3}} = 0{,}006$
Span drift	$u(Corr_{\text{s,dr}})$	$\frac{(0{,}5/100)\text{x}\,100}{\sqrt{3}} = 0{,}29$
Sensitivity to the sample volume flow	$u(Corr_{\text{s,vf}})$	$\frac{(0{,}2/100)}{10}\text{x}\,100\,\text{x}\frac{5\times 2}{\sqrt{3}} = 0{,}12$
Sensitivity to atmospheric pressure	$u(Corr_{\text{a,press}})$	$(0{,}4/100)\text{ x }40 \times \frac{100 - 99}{\sqrt{3}} = 0{,}09$ for CO
Sensitivity to ambient temperature	$u(Corr_{\text{temp}})$	$\frac{(0{,}5/100)}{10}\text{x}\,100\,\text{x}\sqrt{\frac{(308-285)^2 + (308-285)(283-285) + (283-285)^2}{3}} = 0{,}64$
Sensitivity to electric voltage	$u(Corr_{\text{V}})$	$\frac{(0{,}12/100)}{10}\text{x}\,100 \times \frac{2\times(230/100\times 5)}{\sqrt{12}} = 0{,}08$
Interferent: CH_4	$u(Corr_{\text{CH4}})$	$\frac{1{,}6}{50}\text{x}\sqrt{\frac{10^2}{3}} = 0{,}19$
Interferent: CO_2	$u(Corr_{\text{CO2}})$	$\frac{-0{,}6}{15}\text{x}\sqrt{\frac{12^2 + 12\times 8 + 8^2}{3}} = -0{,}41$
Repeatability in laboratory at span level	$u(Corr_{\text{rep}})$	$(0{,}45/100)\text{ x }100 = 0{,}45$
Uncertainty of calibration gas	$u(Corr_{\text{adj}})$	$\frac{(2/100)\text{ x }40}{2} = 0{,}4$

B.5.5 Calculation of combined uncertainty

$$S_{\text{int,n}} > S_{\text{int,p}}$$

$$u(C_{\text{CO,ppm}}) = \sqrt{\begin{aligned} & u^2(Corr_{\text{fit}}) + u^2(Corr_{\text{0,dr}}) + u^2(Corr_{\text{s,dr}}) + u^2(Corr_{\text{rep}}) + u^2(Corr_{\text{adj}}) + u^2(Corr_{\text{s,vf}}) \\ & + u^2(Corr_{\text{a,press}}) + u^2(Corr_{\text{temp}}) + u^2(Corr_{\text{volt}}) + S_{\text{int,n}}^2 \end{aligned}}$$

$$u(C_{\text{CO,ppm}}) = 1{,}12 \text{ ppm}$$

B.6 Conversion of the concentrations in mg/m³(n)

Concentration in mg/m^3(n) is calculated according to the equation:

$$C_{\text{mg/m}^3} = C_{\text{ppm}} \frac{M_{\text{mol}}}{V_{\text{mol,std}}}$$

If the uncertainties associated to M_{mol} and $V_{\text{mol,std}}$ are neglected (uncertainties are due to the fact that values are rounded and depend of the number of digits), standard uncertainty associated to the result calculated in mg/m^3(n) is given by:

$$u^2(C_{\text{mg/m}^3(n)}) = \left(\frac{M_{\text{mol}}}{V_{\text{mol,std}}}\right)^2 u^2(C_{\text{ppm}})$$

$$u(C_{\text{CO,mg/m}^3}) = 1{,}40 \text{ mg/m}^3\text{(n)}$$

$$U_{\text{c}}(C_{\text{CO,mg/m}^3(n)} = \pm 2{,}8 \text{ mg/m}^3 \text{ (n) } (k = 2)$$

$$U_{\text{c,rel}}(C_{\text{CO,mg/m}^3(n)\text{,rel}}) = \pm 5{,}6 \text{ \% relative } (k = 2)$$

B.7 Evaluation of the compliance with the required measurement quality

Interferents with a positive impact:

$$\text{Total} = \frac{1{,}6 + 0{,}8}{100} \times 100 = 2{,}4 \text{ \%} \quad \text{of the range}$$

Total < 4 % of the range

Interferents with a negative impact:

$$\text{Total} = \frac{-1{,}6}{100} \times 100 = -1{,}6\ \% \quad \text{of the range}$$

$$\left|\text{Total}\right| < 4\ \%\ \text{of the range}$$

All values of the performance characteristics obtained in tests are complying with the requirements.

Conclusion: the measurement method fulfils the requirements.

Annex C
(informative)

Evaluation of the method in the field

C.1 General

The method has been evaluated during six field tests, on waste incineration installations, co-incineration installations and large combustion plants. Each test was performed by at least four different European measuring teams originating from ten member countries.

The method has been evaluated during 6 field tests, on waste incineration installations, co-incineration installations and large combustion plants.

C.2 Characteristics of installations

— 1st field test: INERIS bench-loop at Verneuil en Halatte (F); the bench-loop simulates combustion or waste incineration exhaust gases. Five teams took part in the 1st field test. Double measurements were not performed simultaneously but sequentially. Five different flue gas matrices were generated. Within each matrix, two sequential measurements were performed. Two additional sequential measurements were performed in flue gas matrices where the flue gas concentrations varied. There were a total of twelve measurements performed by all the teams.

— 2nd field test: waste incinerator in Denmark. Four teams took part to the field test and performed double measurements simultaneously. A total of sixteen measurements were performed by all the teams.

— 3rd field test: waste incinerator in Italy. Four teams took part to the field test. Two pairs of two teams performed double measurements simultaneously and the four teams performed single measurements simultaneously. A total of six double measurements were performed by each pair of two teams while a total of twelve single measurements were performed by all teams.

— 4th field test: co-incinerator combined heat and power installation in Sweden. The fluidised bed boilers operate on fuel mixes of wood chips, demolition waste, peat and coal. Two pairs of two teams performed double measurements simultaneously and the four teams performed single measurements simultaneously. A total of six double measurements were performed by each pair of two teams while a total of twelve single measurements were performed by all the teams.

— 5th field test: co-incinerator cement plant in Germany. The fuel could be coal, heavy oil and secondary fuel (e.g. paper, plastics, textiles, and tires). Four teams took part to the field test and performed doubles measurements simultaneously. All the teams performed a total of sixteen double measurements.

— 6th field test: coal fired power plant in Germany. Four teams performed their double measurements simultaneously. The total amount of double measurements performed by all teams is 12.

An overview of the flue gas characteristics is given in Table C.1.

Table C.1 — Flue gas characteristics during field tests

Field test	Installation	Fuel	Flue gas characteristics						
			T °C	O_2 % vol	NOx mg/m^3	SO_2 mg/m^3	CO mg/m^3	H_2O % vol	PM mg/m^3
1	Power plant[a]	Natural gas	< 150	5 to 13	10 to 1 300	10 to 2 000	20 to 400	10 to 21	<1
2	Waste incin-erator	Municipal waste	90 to 110	8 to 11	180 to 250	25 to 250	5 to 15	13 to 19	1 to 5
3	Waste incin-erator	Municipal waste	85 to 105	16 to 18	61 to 78	5 to 50	0 to 2	8 to 12	1 to 5
4	Co-incinerator	Wood, waste, coal	70 to 80	4 to 6	4 to 70	0 to 10	50 to 150	8 to 12	0 to 20
5	Co-incinerator	Coal, oil, waste	140 to 170	4 to 6	440 to 1060	60 to 170	260 to 740	23 to 26	5 to 10
6	Power plant	Coal	130 to 140	8,9 to 9,2	110 to 140	1000 to 1130	3 to 6	5,5 to 8	<50

[a] bench-loop: flue gas simulation.

C.3 Repeatability and reproducibility in the field

Repeatability standard deviation s_r and reproducibility standard deviation s_R are determined by performing inter-laboratory tests.

Repeatability standard deviation s_r, internal confidence interval (CI_r) and repeatability in the field r are calculated according to ISO 5725-2 and ISO 5725-6, from the results of the double measurements implemented by the same laboratory.

$$CI_r = t_{0,95;n-1} \cdot s_r \quad \text{and} \quad r = \sqrt{2} \cdot t_{0,95;n-1} \cdot s_r$$

Where

CI_r is the internal confidence interval;

s_r is the repeatability standard deviation;

$t_{0,95;n-1}$ is the student factor for a level of confidence of 95 % and a degree of freedom of *n*-1 (*n*: number of double measurements);

r is the repeatability in the field.

Reproducibility standard deviation s_R, external confidence interval (CI_R) and reproducibility in the field R are calculated according to ISO 5725-2 and ISO 5725-6, from the results of parallel measurements performed simultaneously by several laboratories.

$$CI_R = t_{0,95;np-1} \cdot s_R \quad \text{and} \quad R = \sqrt{2} \cdot t_{0,95;np-1} \cdot s_R$$

Where

CI_R is the external confidence interval;

s_R is the repeatability standard deviation;

$t_{0,95;np-1}$ is the student factor for a level of confidence of 95 % and a degree of freedom of np - 1 (n: number of measurements; p: number of laboratories);

R is the reproducibility in the field.

Table C.2 — Repeatability in the field

Field-test	Concentration		Number of teams	Number of double meas-urements s	Repeatability standard de-viation s_r	95 % confidence interval for a single measurement CI_r	Repeatability r	
	Range mg/m^3	Average mg/m^3					mg/m^3	%
1A	16 to 20	18	5	2	0,24	0,54	0,76	4,3
1B	57 to 69	64	5	2	0,34	0,76	1,08	1,7
1C	95 to 106	100	5	2	0,42	0,94	1,33	1,3
1D	205 to 211	209	5	2	0,65	1,5	2,08	1,0
1E	338 to 353	345	5	2	0,95	2,1	3,03	0,9
1F	213 to 278	256	5	1	0,75	2,1	2,95	1,2
1G	41 to 53	44	5	1	0,30	0,82	1,16	2,6
2	7,2 to 15	9.5	4	16	2,7	5,9	8,29	86,9
3	<dl							
4	95 to 183	144	4	12	2,9	6,5	9,19	6,4
5	264 to 441	328	3	12	4,8	11	15,55	4.7
6	3,8 to 5,6	4,5	4	12	1,2	2,7	3,82	84,8

The following functions were determined:

$s_r(C)$= 0,0088 C + 1,83 in mg/m^3;

$s_{r\ limit}(C)$= 0,0118 C + 2,94 in mg/m^3;

$CI_r(C)$= 0,020 C + 4 in mg/m^3;

$r(C)$= 0,0283 C + 5,7 in mg/m^3.

Table C.3 — Reproducibility in the field

Field test	Concentration		Number of teams	Number of measurements	Reproducibility standard deviation s_R	95% confidence interval for a single measurement CI_R	Reproducibility R	
	Range mg/m^3	Average mg/m^3					mg/m^3	%
1A	16 to 20	18	5	2	1,3	3,0	4,2	23
1B	57 to 69	64	5	2	2,2	4,9	7,0	11
1C	95 to 106	100	5	2	1,8	4,0	5,7	6
1D	205 to 211	209	5	2	2,4	5,4	7,7	4
1E	338 to 353	345	5	2	2,5	5,7	8,1	2
1F	213 to 278	256	5	1	23	64	90,3	35
1G	41 to 53	44	5	1	4,8	13	18,7	42
2	7,2 to 14,3	9,5	4	16	3,5	7,6	10,7	112
3	<dl							
4	57 to 149	93	4	12	4,8	11	15,6	17
5	264 to 441	328	3	12	6,6	15	21,2	6
6	3,8 to 5,6	4,5	4	12	2,0	4,5	6,4	141

The following functions were determined:

$S_R(C)$= 0,0118 C + 2,94 in mg/m^3;

$CI_R(C)$=0,029 C + 7,2 in mg/m^3;

$R(C)$= 0,041 C + 10,2 in mg/m^3.

Annex D
(informative)

Procedure of correction of data from drift effect

The drift is supposed to be proportional to the time

Adjustment and span check after measurement

Gas	expected value	reading before	reading after
Span	918.9	912.3	882.4
zero	0	1.8	1.7
Time		5:36	17:58
duration		F7-E7	
in minutes		742	

Fill in only boxes with shaded area

Relation between the true value (Y) and the reading (X), Y=AX+B

Coefficients	Adjustment	Check	deviation	Drift / min.
A (span)	(D5-D6)/(E5-E6)	(D5-D6)/(F5-F6)	E13-D13	(E13-D13)/E9
B (zero)	D13*-E6	E13*-F6	E14-D14	(E14-D14)/E9

Drift is ***1-((D5-(D5*F13+F14))/D5)***

Equation to calculate the corrected concentration (Ccor.) from the reading (C) and time (t)

C cor. = C x (C14 + E14 x t) + (C15 + E15 x t)

for this application :

CONCATENER("C cor. = C x (";CTXT(D13;4);" + ";CTXT(G13;4);" x t) + (";CTXT(D14;4);" + ";CTXT(G14;4);" x t)")

Exemple

C	Time	C cor.
E5	0	D28*(D$13+G$13*E28)+(D$14+G$14*E28)
F5	E9	D29*(D$13+G$13*E29)+(D$14+G$14*E29)
180	124	D30*(D$13+G$13*E30)+(D$14+G$14*E30)

Annex E
(informative)

Relationship with EU Directives

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association and supports essential requirements of the EU Council Directive 2000/76/EC on the Incineration of Hazardous Waste and of the EU Council Directive 2001/80/EC on the limitation of emissions of certain pollutants into the air from large combustion plants.

WARNING — Other requirements and EU Directives may be applicable falling within the scope of use of this European Standard.

Bibliography

[1] Council Directive 2001/80/EC on the limitation of emissions of certain pollutants into the air from large combustion plants

[2] Council Directive 2000/76/EC on waste incineration plants

[3] EN 14790:2005, *Stationary source emissions – Determination of the water vapour in ducts*.

[4] EN 13284-1:2001, *Stationary source emissions – Determination of low range mass concentration of dust – Part 1: Manual gravimetric method*.

[5] ISO 9169, *Air quality* – Determination of performance characteristics of measurement methods.

[6] ISO 11095:1996, *Linear calibration using reference materials*.

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